



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
CS&IT DEPARTMENT
PROGRAM BSCS

COURSE INFORMATION SHEET
(For Lab Based Course)

Session:	Fall-2024
Course Title:	Web Engineering
Course Code:	CS-311L
Credit Hours:	0+1
Semester:	5 th
Pre-Requisites:	PROGRAMMING FUNDAMENTALS (CS-116L)
Instructor Name:	Mr. Muhammad Rahil Usman
Email and Contact Information:	rausman@ssuet.edu.pk
WhatsApp Group	WebEngineering2024
Office Hours:	Monday to Friday (09:00 AM to 05:00 PM) ✓
Mode of Teaching:	Synchronous/Asynchronous/ Hybrid/Blended

Course Objectives:

Upon successful completion of the course, the student will demonstrate the ability to:

- Understand the major areas and challenges of web programming.
- Distinguish web-related technologies.
- Be able to build ANY website you want
- Build fully-fledged websites and web apps for your startup or business
- Understand the latest frameworks and technologies, including Bootstrap 4, Mongo DB, NodeJS, Express.

Course Description:

Web Design: HTML 5, CSS 3. Responsive Web Design: Bootstrap 5. Front-End Web Development: JavaScript ES6, DOM Manipulation, JQuery, Bash Command Line, Git, GitHub and Version Control. Backend Web Development: Node.js, NPM, Express.js, REST, APIs. Database: MongoDB. Deployment: with GitHub Pages, Heroku, and Mongo DB Atlas



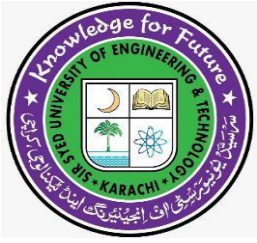
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COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes(PLOs):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Describe Web standards, web applications Modern protocols, programming concepts and control structures used in web programming.	1	Remember
2	Explain Programs having HTML,CSS,JavaScript protocols standards.	1	Understand
3	Apply Apply knowledge and understanding to solve interactive web development problems and project their solutions under specific requirements.	3	Apply

GRADING POLICY:

Assessment Tools	Percentage	Marks
Lab Manual	30%	15
Viva/Demonstration	30%	15
Lab Exam/Project	40%	20
TOTAL	100%	50



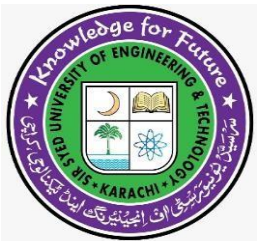
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Recommended Book:

- Jon Duckett HTML and CSS Design and Build Websites
- ***FREEMAN, ADAM THE DEFINITIVE GUIDE TO HTML5:***
- Ladwa, Hanumanth HTML, CSS, and JavaScript
- Terry Ann Felke-Morris, Ed.D Web Design HTML5 & CSS
- A Smarter Way to Learn JavaScript

Reference Books:

- Emilia Mendes and Nile Mosley, Springer Verlag Web Engineering, , 2010.
- Roger S. Pressman, McGraw Hill, Web Engineering: A Practitioners' Approach, 2008.
- Dynamic HTML: The Definitive Reference: A Comprehensive Resource for XHTML, CSS, DOM, JavaScript O'Reilly Media 3rd Edition, 2007.
- David Flanagan. O'Reilly Media JavaScript: The Definitive Guide, 8th Edition,.
- Dreamtech Press Black Book Web Technologies, ,
- P.J. Deitel & H.M. Deitel Pearson. Internet and World Wide Web How to program,



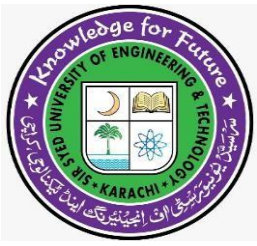
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LAB PLAN

Course Title: Web Engineering
Course Code: CS-311

Week #		Topics	Required Reading
1	14-10-2024 to 18-10-2024	Client-side technologies: HTML-I • The Anatomy of an HTML Tag • What we're building - HTML Personal Site • What is The HTML Boilerplate? • How to Structure Text in HTML • HTML Lists • HTML Image Elements • HTML Links and Anchor Tags	Lab Manual
2	21-10-2024 to 25-10-2024	Client-side technologies: HTML-II • HTML Tables • Using HTML Tables for Layout • HTML Tables Code Challenge • How to Type Emojis • HTML Forms	
3	28-10-2024 to 01-11-2024	Client-side technologies: CSS - I • Introduction to CSS • The Anatomy of CSS Syntax • Inline CSS, Internal CSS and External CSS • CSS Selectors • Classes vs. Ids	
4	04-11-2024 to 08-11-2024	Client-side technologies: CSS - II • What Are Favicons? • Divs • The Box Model of Website Styling • CSS Display Property • CSS Static, Relative and Absolute positioning • CSS Typography	Quiz

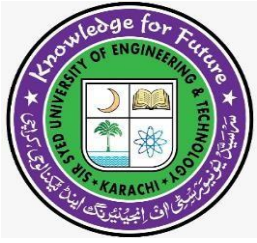


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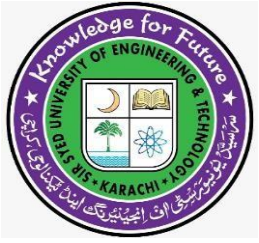
5	11-11-2024 to 15-11-2024	Client-side technologies: Bootstrap What is Bootstrap? • Installing Bootstrap • Web Design - Wire framing • The Bootstrap Navigation Bar • Bootstrap Grid Layout System • Adding Grid Layouts to Our Website • Bootstrap Containers • Bootstrap Buttons & Font Awesome	Assignment
6	18-11-2024 to 22-11-2024	• Client-side technologies: Javascript - I • Introduction to JavaScript • JavaScript Alerts - Adding Behavior to Websites • Data Types • JavaScript Variables • String Functions • Basic Arithmetic and the Modulo Operator in JavaScript • Increment and Decrement Expressions • Control Statements: Using If-Else Conditionals & Logic • Control Statements: While Loops • Control Statements: For Loops • JavaScript Objects and Functions	
7	25-11-2024 to 29-11-2024	Client-side technologies: Javascript - II	
8	MID TERM EXAMINATION (02-12-2024 to 07-12-2024)		
9	09-12-2024 to 13-12-2024	Client-side technologies: jQuery and JSON	
10	16-12-2024 to 20-12-2024	Client-side technologies: AJAX and React JS • Learn the purpose of Asynchronous JavaScript And XML (AJAX)	
11	23-12-2024 to 27-12-2024	Server-side technologies: C#, WebAPI Basics	
12	30-12-2024 to 03-01-2025	Server-side technologies: LINQ queries, MVC Framework, C# Data First approach	Quiz



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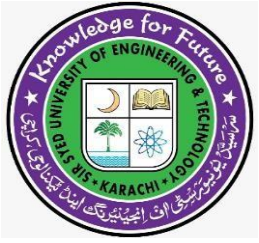
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13	06-01-2025 to 10-01-2025	Server-side technologies: C# Data First Approach with Web APIs – Web API Consumption	Assignment
14	13-01-2025 to 17-01-2025	Deployment, Operation & Maintenance of Web Applications	
15	Final Exam		



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